

Notes: Hortacsu and Syverson (QJE, 2004)
Product Differentiation, Search Costs, and Competition in the Mutual Fund
Industry: A Case Study of S&P 500 Index Funds

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1 Big picture

- **Question.** Why does the mutual fund industry have so many funds and such large fee dispersion, even within very narrow sectors where portfolios should be nearly identical?
- **Case study.** The paper studies retail S&P 500 index funds. This is a particularly clean setting because all of these funds explicitly seek to replicate the same benchmark. If strong fee dispersion and entry still appear here, then something outside pure portfolio choice must be going on.
- **Puzzle.** In 2000 there were 85 retail S&P 500 index funds. The cheapest fund charged about 9.5 basis points and the most expensive charged about 268 basis points, yet the highest-fee fund was still the tenth-largest in the sector. So this is not a story in which expensive funds are all tiny fringe products.
- **Main candidate mechanisms.**
 - *Nonportfolio differentiation:* investors may care about things like fund age, family reputation, exchange-traded status, bundled advice, or tax features.
 - *Search and information frictions:* investors may not find or compare all funds, so they do not all end up in the cheapest or highest-utility option.
 - *Switching costs:* investors may be partially locked into fund families, especially when loads or family-level relationships matter.
- **What the paper does.** It combines descriptive evidence with a structural search model of mutual fund demand and pricing. Investors search over funds with heterogeneous search costs, and funds may either differ in their visibility to investors or differ in nonprice attributes investors value.
- **Main findings.**
 - Observable extra-portfolio fund attributes matter for demand.
 - Search frictions can rationalize why the highest-utility fund does not capture the entire market.
 - Search costs fall for much of the investor distribution over the late 1990s, but rise in the upper tail.
 - This upper-tail rise appears concentrated among load-fund buyers, which the authors interpret as consistent with an influx of novice, high-information-cost investors.
- **Welfare angle.** Too many nearly identical funds can be socially costly because they induce wasteful search and may sacrifice scale economies. But those losses must be balanced against product-variety gains and any pro-competitive effect of entry.
- **Broader contribution.** The paper is about more than mutual funds. It shows how one can use aggregate price and market-share data to learn about search frictions in a differentiated-product industry, even without investor-level choice data.
- **Takeaway.** The mutual fund market is not just a contest of portfolio returns. Competition is also shaped by investor information frictions and by nonportfolio product differentiation, even in a sector where financial performance is almost homogeneous.

2 Data and measurement (key objects)

2.1 Data sources and sample

The main data come from the CRSP mutual fund database for 1995–2000, supplemented with fund cost data from Lipper Analytical Services. CRSP provides returns, assets, manager identity, management company, age, loads, expense ratios, 12b-1 fees, and other fund characteristics. Lipper provides category-level cost information based on year-end reports.

The core empirical sample is the **retail S&P 500 index fund sector**. The authors:

- include noninstitutional S&P 500 index funds,
- treat different share classes as separate funds,
- include exchange-traded funds such as SPDRs (and later iShares),
- exclude institutional funds,
- and exclude “enhanced” index funds that combine passive and active strategies.

For the structural estimation, the sample consists of all retail S&P 500 funds operating in any year from 1995 to 2000, excluding ten fund-year observations lacking fee or asset data. This leaves **309 fund-year observations**.

2.2 Why this sector is unusually clean

The attraction of the setting is that these funds are supposed to deliver the same underlying asset exposure. The paper shows that realized performance differences are indeed small:

- the interquartile range of annual gross returns is never larger than about **0.32 percentage points**,
- and dispersion in monthly return volatility is also tiny.

This means large fee differences are hard to explain by saying that some funds simply offer much better portfolios. The case study deliberately strips away much of the usual cross-fund portfolio heterogeneity.

2.3 Price measure

The main price variable is the literature-standard annualized fee measure:

$$\text{Price} = \text{expense ratio} + 12\text{b-1 fee} + \frac{1}{7} \times \text{total loads.} \quad (1)$$

Everything is expressed in **basis points**. The one-seventh weight on loads reflects the usual assumption that investors hold fund shares for about seven years on average.

The paper also checks robustness using a narrower price definition that ignores loads and uses **annual fees only**. This matters because load incidence depends on investor holding period, especially for back-end and deferred-load funds.

2.4 Market share measure

The benchmark market share is **asset-based**: a fund's share of total sector assets. This is natural because assets under management are observed directly and because fund fees are largely charged on asset stocks.

The authors also build an alternative **flow-based** market-share proxy by summing positive monthly net inflows through the year. They note that this proxy is highly correlated with the asset-based measure, with a correlation around **0.93**.

2.5 Main nonprice fund attributes

In the differentiated-products specification, the paper considers several fund attributes that may matter to investors beyond price itself:

- whether the fund has **any load**,
- whether it has a **rear or deferred load**,
- whether it is an **exchange-traded fund (ETF)**,
- the **number of other share classes** tied to the same portfolio,
- the log of the **number of funds in the same management company**,
- the log of **fund age**,
- **manager tenure**,
- the **income plus capital gains yield** (a tax-exposure proxy),
- the average monthly difference between the fund's return and the **S&P 500 return**,
- and the **standard deviation of monthly returns**.

2.6 Marketplace exposure and sampling probability

In the unequal-sampling-probability version of the model, investors are more likely to discover some funds than others. The paper uses **fund age** as the benchmark visibility variable. In an extension, visibility depends on both age and the size of the fund family.

Interpretation. An older fund or a fund belonging to a larger family may be easier to find because of reputation, past investor exposure, or indirect familiarity from other family products.

2.7 Descriptive market evolution

Table II documents several important changes in the sector from 1995 to 2000:

- the number of funds with price data rises from **24** to **82**,
- mean price rises from **82.4** to **97.1** basis points,
- the asset-weighted mean rises from **26.8** to **32.2** basis points,

- the largest fund’s share falls from **78.9%** to **53.9%**,
- the Herfindahl index falls from **6281** to **3208**,
- the low-price decile’s share falls from **86.0%** to **74.9%**,
- and the highest-price quartile’s share rises from **1.4%** to **4.1%**.

Interpretation. Entry is substantial and concentration declines, but this does *not* produce a flight toward the cheapest funds. Instead, sector assets shift noticeably toward more expensive products.

2.8 Institutional funds as a comparison group

The paper also examines institutional S&P 500 index funds. Their prices are lower and their price distribution is tighter than those of retail funds. The authors view this as supportive evidence for the importance of search or information frictions: institutional investors likely have lower search costs and stronger incentives to compare fees.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper is organized around three linked ideas.

Hypothesis 1: search frictions can sustain fee dispersion. Even if funds are financially similar, investors may fail to locate the cheapest or highest-utility option.

Hypothesis 2: extra-portfolio differentiation matters. Investors may value fund age, family size, ETF status, tax features, and other attributes that are not simply the portfolio return process.

Hypothesis 3: late-1990s asset shifts reflect changing investor composition. If novice investors enter the market, the upper tail of the search-cost distribution may rise, especially among buyers of load funds who value broker advice.

3.2 Investor utility from a fund

The fund-level indirect utility is

$$u_j = W_j' \beta - p_j + \xi_j, \tag{2}$$

where W_j are observed nonprice attributes, p_j is price, and ξ_j is an unobserved fund attribute.

Interpretation. Price enters with coefficient -1 , so utility is measured in basis points. A positive coefficient on an attribute means investors are willing to pay more for funds with more of that attribute.

3.3 Optimal search rule

An investor with search cost c_i continues to search as long as the expected benefit exceeds the cost:

$$c_i \leq \int_{u^*}^{\bar{u}} (u - u^*) dH(u), \quad (3)$$

where u^* is the best utility found so far and $H(u)$ is the investor's belief about the distribution of fund utilities.

With equal sampling probabilities, investors take the empirical utility distribution as given:

$$H(u) = \frac{1}{N} \sum_{j=1}^N I(u_j \leq u). \quad (4)$$

Interpretation. Search stops when the expected gain from one more draw is smaller than the investor's search cost.

3.4 Critical search-cost cutoffs

Label funds so that $u_1 < \dots < u_N$. Then the cutoff search cost for buying fund j is

$$c_j = \sum_{k=j}^N \rho_k (u_k - u_j), \quad (5)$$

where ρ_k is the probability that fund k is sampled on any search.

Interpretation. A fund with low utility has a high cutoff search cost, because any investor who lands there still expects to gain a lot from continuing to search. The best fund has $c_N = 0$.

3.5 Market-share equations

For the lowest-utility fund,

$$q_1 = \rho_1 (1 - G(c_1)). \quad (6)$$

For the second-lowest-utility fund,

$$q_2 = \rho_2 (1 - G(c_1)) + \frac{\rho_2}{1 - \rho_1} [G(c_1) - G(c_2)]. \quad (7)$$

More generally, for $j \geq 3$,

$$q_j = \rho_j \left[1 + \frac{\rho_1 G(c_1)}{1 - \rho_1} + \frac{\rho_2 G(c_2)}{(1 - \rho_1)(1 - \rho_1 - \rho_2)} + \sum_{k=3}^{j-1} \frac{\rho_k G(c_k)}{\left(1 - \sum_{\ell=1}^{k-1} \rho_\ell\right) \left(1 - \sum_{\ell=1}^k \rho_\ell\right)} \right. \quad (8)$$

$$\left. - \frac{G(c_j)}{1 - \sum_{\ell=1}^{j-1} \rho_\ell} \right]. \quad (9)$$

Interpretation. Market shares depend on two objects: the probability a fund is encountered and

the distribution of search costs. This is the heart of the model.

3.6 Fund supply and pricing

Fund j chooses price to maximize profits:

$$\Pi_j = Sq_j(p, W)(p_j - mc_j), \quad (10)$$

where S is total market size and mc_j is marginal cost.

The first-order condition is

$$q_j(p, W) + (p_j - mc_j) \frac{\partial q_j(p, W)}{\partial p_j} = 0. \quad (11)$$

The paper derives a closed-form expression for $\partial q_j / \partial p_j$ as a weighted sum of search-cost densities $g(c_k)$ evaluated at the critical cutoffs.

Interpretation. Markups are pinned down by how many investors sit on each search margin. Search frictions therefore affect prices directly through demand elasticities.

3.7 Homogeneous-product special case

If funds differ only in price, then $u_j = u' - p_j$ and the search cutoff simplifies to

$$c_j = \sum_{k=j}^N \rho_k (p_j - p_k). \quad (12)$$

Interpretation. In the homogeneous case, prices and sampling probabilities are enough to recover search cutoffs directly.

3.8 Recovering the search-cost distribution from pricing

Under Bertrand-Nash pricing,

$$\frac{\partial q_j(p)}{\partial p_j} = -\frac{q_j(p)}{p_j - mc_j}. \quad (13)$$

Once the paper has the c.d.f. and the density of search costs at the relevant cutoffs, it recovers the spacing between those cutoffs using a trapezoid approximation:

$$G(c_{j-1}) - G(c_j) = 0.5 [g(c_{j-1}) + g(c_j)](c_{j-1} - c_j), \quad (14)$$

so that

$$c_{j-1} - c_j = \frac{2 [G(c_{j-1}) - G(c_j)]}{g(c_{j-1}) + g(c_j)}. \quad (15)$$

Interpretation. The paper uses observed prices and shares, together with firms' pricing optimality conditions, to back out the search-cost distribution.

3.9 Utility regression for differentiated funds

After recovering indirect utilities, the paper estimates how observed attributes contribute to utility using

$$u_j + p_j = X_j\beta + \beta_{age} \ln(age_j) + \eta_j. \quad (16)$$

Interpretation. This translates recovered fund utilities into economically interpretable willingness-to-pay coefficients for observed fund characteristics.

3.10 Unequal sampling probabilities

To allow some funds to be easier to find, the paper specifies

$$\rho_j = \frac{Z_j^\alpha}{\sum_{k=1}^N Z_k^\alpha}, \quad (17)$$

where Z_j is an exposure measure such as age.

Interpretation. If $\alpha > 0$, funds with greater visibility are disproportionately likely to be sampled by investors.

3.11 Estimation strategy

The estimation proceeds in layers.

- First, the simplest homogeneous-product/equal-sampling model is checked and rejected by the data.
- Next, the paper estimates a homogeneous-product model with unequal sampling probabilities. Search costs are parameterized as lognormal, and the model is fit by nonlinear least squares to prices and market shares.
- Finally, the paper estimates a differentiated-products model with equal sampling probabilities, recovers search costs nonparametrically, and then estimates utility loadings by IV.

The paper also compares asset-based versus flow-based market shares and load-inclusive versus annual-fee-only price measures.

4 Identification and credibility (what makes the model persuasive)

4.1 This is not an IV paper

The paper does not identify causal effects through an external shock or natural experiment. Its credibility comes from a different source: a carefully chosen market setting and a tightly structured equilibrium model.

Core standard. The model should be able to explain prices and market shares in a market where

portfolio fundamentals are almost constant across products.

4.2 Why the S&P 500 index-fund sector is so useful

This paper's central empirical advantage is the setting itself. By focusing on retail S&P 500 index funds, the authors eliminate much of the usual within-sector variation in expected returns and risk.

That is why the pricing patterns are so striking. If very similar products still exhibit large price spreads and many firms, then search and nonportfolio differentiation become first-order explanations rather than side issues.

4.3 What is identified from prices and shares

In the homogeneous-products case, once the sampling probabilities are known or parameterized, prices and market shares identify the relevant search-cost cutoffs.

In the differentiated-products case, market shares identify the c.d.f. of search costs only up to cutoff locations. Pricing first-order conditions are then used to recover the relevant densities and, from there, the spacing between cutoffs. This is where the supply side matters.

4.4 Why Figure III is an important rejection test

The simplest version of the model predicts a monotone negative relationship between price and market share. Figure III shows that the data do not look like that: while price and market share are negatively correlated, the relationship is far from monotone.

Why this matters. The paper does not simply assume differentiation or visibility effects. It first shows why a pure price-only story is inadequate.

4.5 Why age must be instrumented

In the utility regression, fund age can be correlated with unobserved quality. Older surviving funds may have higher latent quality, so OLS would confound observed age effects with survival-based selection.

The paper therefore instruments for logged age using Berry-Levinsohn-Pakes style instruments built from current-year attributes of:

- the fund itself,
- other funds in the same family,
- and funds outside the family.

These instruments vary by fund-year and are meant to capture how a fund's position in attribute space affects market structure independently of unobserved quality.

4.6 What the paper cannot identify separately

A crucial limitation is that with only aggregate prices and shares, the paper cannot separately identify:

- unequal sampling probabilities and nonprice differentiation at the same time,
- search costs and switching costs cleanly,
- or general horizontal taste heterogeneity apart from search frictions.

This is why the paper studies unequal sampling and differentiated products in separate specifications rather than in one fully unrestricted model.

4.7 Search costs versus switching costs

The authors take switching costs seriously but do not build them explicitly into the baseline model. They provide two simple empirical checks:

- whether S&P 500 fund flows in load families react differently to the performance of other family funds,
- and whether within-family S&P 500 shares grow faster in load families.

They do not find strong evidence for these patterns. That does not prove switching costs are absent, but it suggests that switching costs are probably not the dominant driver of the sector's pricing patterns.

4.8 Supportive evidence for search frictions

Several pieces of evidence support the search interpretation.

- Retail funds have much more price dispersion than institutional S&P 500 funds.
- Surveys show investors consult multiple information sources before buying funds.
- The market share shift toward expensive funds coincides with the entry of many novice mutual-fund investors in the late 1990s.

These are not identification devices in the experimental sense, but together they make the structural interpretation more persuasive.

4.9 Robustness and internal checks

The paper checks several alternative empirical choices.

- asset-based versus flow-based market shares,
- load-inclusive versus annual-fee-only prices,
- sampling probability depending on age alone versus age plus family size,
- and differentiated-products calculations under alternative assumed marginal costs.

The qualitative conclusions survive these changes.

4.10 Best-response verification

Because the differentiated-products model uses firms' pricing first-order conditions, the paper checks whether estimated demand parameters imply that observed prices are close to the firms' best responses. The authors report that the profit functions are single-peaked and attain maxima near the observed prices.

Why this matters. Structural pricing equations are only persuasive if the estimated demand system is consistent with firms actually wanting to set prices near what we observe.

4.11 Main limitations

The paper is candid about what it cannot do.

- It does not have investor-level data.
- It cannot separately estimate a fully general model with both search frictions and broad taste heterogeneity.
- The welfare exercise is stylized and depends on additional modeling assumptions.

These are real limitations, but they do not overturn the paper's main point that extra-portfolio differentiation and investor frictions matter in this market.

5 Tables and figures

5.1 Table I: within-sector fee dispersion is huge even in the index-fund sector

Read:

- In 2000, retail S&P 500 index funds have a mean price of **97.1** basis points.
- The coefficient of variation is **0.677**.
- The 75th-to-25th percentile price ratio is **3.1**.
- The 90th-to-10th percentile ratio is **8.2**.

Economic message. The S&P 500 sector is not an outlier with tiny price spreads. It is actually at the high end of the cross-sector dispersion distribution. This is the paper's opening fact.

5.2 Table II and Figure I: funds proliferate and assets move toward expensive products

Read:

TABLE I
PRICE DISPERSION WITHIN FUND SECTORS

Sector	N	Mean price	Coefficient of variation	75th to 25th Percentile ratio	90th to 10th Percentile ratio
Aggressive growth	1278	191.0	0.485	2.0	3.1
Balanced growth	472	164.2	0.439	2.2	3.7
High-quality bonds	862	118.1	0.566	2.5	4.9
High-yield bonds	337	167.3	0.387	2.1	3.2
Global bonds	358	182.3	0.402	2.0	3.5
Global equities	452	228.3	0.374	1.6	2.8
Growth and income	978	158.4	0.830	2.5	5.5
Ginnie Mae	182	144.0	0.460	2.4	4.0
Gov't securities	450	131.9	0.549	2.5	4.7
International equities	1267	225.5	0.432	1.9	3.2
Income	218	170.8	0.415	2.2	3.4
Long-term growth	1812	179.4	0.421	2.0	3.1
Tax-free money market	455	62.7	0.440	1.6	3.2
Gov't securities money market	437	59.5	0.611	1.8	4.8
High-quality muni bond	541	137.2	0.624	2.4	4.1
Single-state muni bond	1326	150.3	0.384	1.7	3.6
Taxable money market	541	79.2	0.726	2.0	7.1
High-yield money market	62	160.4	0.408	1.7	3.3
Precious metals	35	256.1	0.399	1.6	3.3
Sector funds	511	200.8	0.364	1.8	2.9
Total return	323	178.2	0.415	1.9	3.3
Utilities	94	182.8	0.359	1.7	3.2
Retail S&P 500 index funds	82	97.1	0.677	3.1	8.2

The table shows measures of price (fee) dispersion among mutual funds in various asset classes in 2000. The fund data used to calculate these figures are from the Center for Research in Security Prices (CRSP). Roughly 95 percent of the funds in the CRSP database are matched to one of these sectors, which are categorized according to the Investment Company Data, Inc. (now Standard and Poor's Microcap) system. We follow the CRSP convention (also common in the literature) of treating each fund share class in multiclass funds as a separate fund. Multiclass funds are those that have a common manager and portfolio, but have different pricing schemes and asset purchase and redemption rules. Prices are computed from CRSP data, and are calculated as the fund's annual fees (both management fees and 12b-1 fees if applicable) plus one-seventh of total loads, assuming a mean holding horizon of seven years, as in Sirri and Tufano [1998]. All prices are expressed in basis points. See Data Appendix for details.

TABLE II
EVOLUTION OF RETAIL S&P 500 INDEX FUND SECTOR

Variable	Statistic	1995	1996	1997	1998	1999	2000
Annual gross return (percent)	N (return data)	23	24	36	54	65	76
	Mean	37.43	22.67	33.24	28.95	20.95	-8.63
	Standard deviation	0.25	1.29	0.19	0.84	0.40	1.23
	Interquartile range	0.31	0.30	0.20	0.26	0.21	0.32
	Mean	1.492	3.133	4.574	6.199	3.808	4.932
	Standard deviation (percent)	0.025	0.038	0.050	0.051	0.115	0.219
Standard deviation of monthly returns (percent)	Interquartile range	0.016	0.023	0.029	0.028	0.024	0.037
	N (price data)	24	33	45	57	68	82
	Minimum	19	18.0	16.0	17.0	17.0	9.45
	25th percentile	43.8	45.0	40.0	46.0	47.1	47.0
	Median	77.5	60.0	70.0	82.0	80.9	72.1
	75th percentile	120.5	123.1	136.3	136.3	152.9	144.8
Price (basis points)	Maximum	206.4	206.4	231.4	231.4	235.4	268.4
	Mean	82.4	80.6	89.8	94.2	104.2	97.1
	Asset-weighted mean	26.8	26.6	26.0	28.9	31.9	32.2
	Standard deviation	50.5	53.1	61.6	60.5	67.1	65.7
	C ₁	78.9	77.0	69.9	62.9	59.9	53.9
	C ₄	89.0	88.4	85.9	82.1	80.1	77.8
Market shares	Herfindahl	6281	5992	5003	4127	3776	3208
	Low-price decile	86.0	84.9	80.8	77.6	75.5	74.9
	High-price quartile	1.4	1.6	1.7	2.5	3.4	4.1

The table shows characteristics of our retail (i.e., noninstitutional) S&P 500 index funds sample. As in Table I, we treat each fund share class in multiclass funds as a separate fund. (If we were to count multiple-class funds as single funds, there would be 50 funds in 2000.) We also include exchange-traded funds (ETFs) based on the S&P 500 index. There is only one ETF for most of our sample, Standard & Poor's Depository Receipts (SPDRs). In 2000 a second ETF, Barclay's iShares S&P 500 Index Fund, started trading. We report price figures for only 82 funds in 2000 because we lack price information for three funds. Reported returns data are limited to funds reporting returns in every month of a given year to avoid comparing figures from funds operating in nonrepresentative portions of the year. C₁ (C₄) is the fraction of sector assets in the largest (largest four) funds. The Herfindahl index is the sum of the squared market shares (expressed as percentages). Low-price decile and high-price quartiles are the combined market shares of the lowest-price and highest-price fund quantiles, respectively.

- The number of funds with price data rises from **24** in 1995 to **82** in 2000.
- The largest fund's share falls from **78.9%** to **53.9%**, and the Herfindahl falls from **6281** to **3208**.
- The low-price decile's market share falls from **86.0%** to **74.9%**.
- The high-price quartile's market share rises from **1.4%** to **4.1%**.
- Figure I shows the entire price distribution shifting right from 1996 to 1999, especially in the upper tail, before reversing somewhat in 2000.

Economic message. Entry and declining concentration do *not* translate into uniform price competition. Instead, the sector expands while expensive funds gain ground.

5.3 Figure II: institutional funds look very different from retail funds

Read: The price histogram for institutional S&P 500 funds is much tighter and centered at lower fees than the retail histogram.

Economic message. Investors with larger stakes and lower information frictions appear much less willing to tolerate large price dispersion. This is exactly the kind of pattern one would expect if search frictions matter.

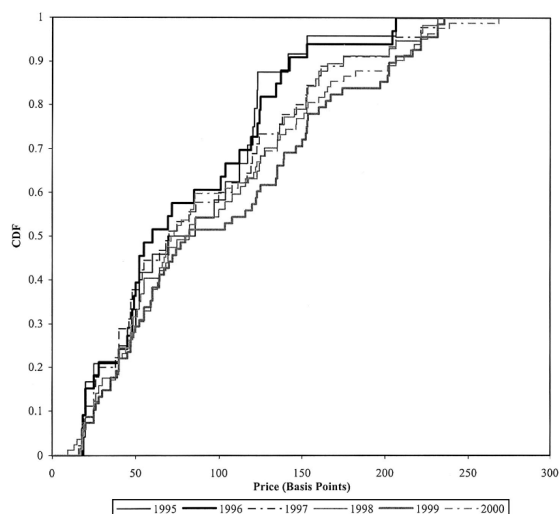


FIGURE I
Price Cumulative Distribution Functions

The figure shows, by year, c.d.f.s of the retail S&P 500 index fund price (fee) distributions. Prices are computed from Center for Research on Security Prices (CRSP) data, and are calculated as the fund's annual fees plus one-seventh of total loads. All prices are expressed in basis points. See Data Appendix for details.

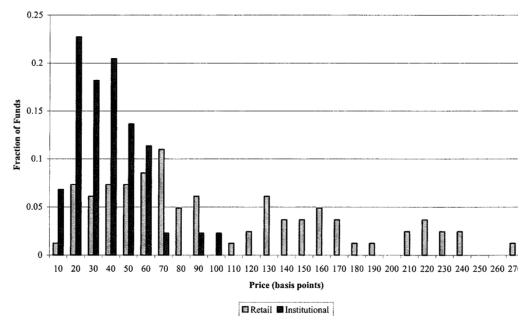


FIGURE II
2000 Price Histograms, Retail and Institutional Funds
The figure shows the price histograms for retail and institutional S&P 500 index funds in 2000. Prices are calculated as annual fees plus one-seventh of total loads. All prices are expressed in basis points.

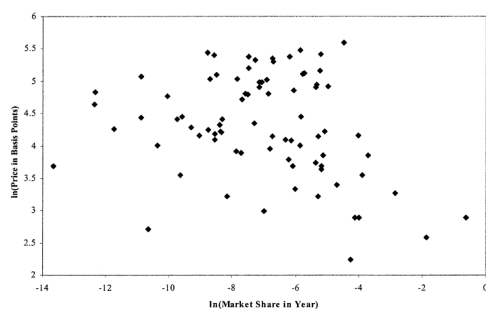


FIGURE III
Log Price Versus Log Market Share, 2000

The figure plots for retail S&P 500 index funds in 2000 the natural logarithm of market share versus the natural logarithm of fund price. Market shares are based on total fund assets, and price are calculated as annual fees plus one-seventh of total loads (expressed in basis points). See Data Appendix for details.

TABLE III
SEARCH MODEL WITH UNEQUAL SAMPLING PROBABILITIES

Parameters	(A)	(B)	(C)	(D)	(E)
Log (mean search cost)	-6.17 (0.06)	-6.68 (0.06)	-6.58 (0.13)	-6.78 (0.26)	-6.33 (0.11)
Variance of logged search costs	1.88 (0.03)	2.07 (0.06)	1.79 (0.06)	1.89 (0.08)	1.95 (0.04)
Mean marginal cost, basis points	4 (1)	12 (1)	11 (2)	16 (2)	4 (1)
α	2.62 (0.04)	2.43 (0.03)	2.58 (0.03)	2.44 (0.09)	3.06 (0.04)
γ	—	—	—	—	0.11 (0.03)
Time trend of mean search cost	-0.38 (0.05)	-0.50 (0.04)	-0.29 (0.08)	-0.51 (0.08)	-0.30 (0.06)
Time trend of search cost variance	0.20 (0.05)	0.18 (0.04)	0.17 (0.02)	0.18 (0.02)	0.16 (0.02)
R^2 , prices	0.92	0.97	0.82	0.94	0.99
R^2 , quantities	0.98	0.83	0.98	0.84	0.98
Median search cost (1996), b.p.	21	12	14	11	18
IQR of search cost (1996), b.p.	5.9 to 75	3 to 50	4 to 46	3 to 41	4.7 to 67
Median search cost (2000), b.p.	5	2	4	1	5
IQR of search cost (2000), b.p.	0.7 to 28	0.3 to 11	0.8 to 23	0.2 to 9	0.9 to 30

This table shows the results from estimating the homogeneous-product/unequal-sampling-probability specification in subsection V.B. Standard errors are in parentheses. b.p. = basis points. The sample consists of all retail S&P 500 funds operating in any year between 1995 and 2000 (inclusive), excluding ten fund-year observations for which we do not have fee or asset data. This leaves a sample of 309 fund-year observations. See text for details.

Key to estimated specifications (see Data Appendix for additional details on market share and price measures):

- (A) Asset-based market shares; prices are annual fees plus one-seventh of total loads.
- (B) Market shares are based on inflows into fund; prices are annual fees plus one-seventh of total loads.
- (C) Asset-based market shares; prices are annual fees only, ignoring loads.
- (D) Market shares are based on inflows into fund; prices are annual fees only, ignoring loads.
- (E) Data as in model (A), but sampling probability is a function of both age and (logged) total number of funds in management company.

5.4 Figure III and Table III: the simple price-only model fails, but visibility plus search works well

Figure III. The log price–log market share relationship in 2000 is negative on average but clearly nonmonotonic. That immediately rejects the simplest equal-sampling homogeneous-products model.

Table III. Read:

- In the benchmark specification, the sampling-probability parameter is $\alpha = 2.62$.
- Estimated mean marginal cost is only about **4 basis points**.
- The model explains about **92%** of price variation and **98%** of quantity variation.
- Median search cost falls from about **21 basis points** in 1996 to about **5 basis points** in 2000.
- The 1996 interquartile search-cost range is roughly **5.9 to 75** basis points, while in 2000 it is about **0.7 to 28**.

Economic message. A remarkably parsimonious search model with unequal visibility can explain most of the pricing and market-share patterns in the data.

5.5 Figure IV and Table IV: product differentiation matters too

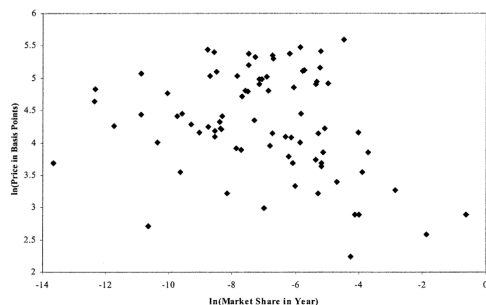


FIGURE III
Log Price Versus Log Market Share, 2000
The figure plots for retail S&P 500 index funds in 2000 the natural logarithm of market share versus the natural logarithm of fund price. Market shares are based on total fund assets, and price are calculated as annual fees plus one-seventh of total loads (expressed in basis points). See Data Appendix for details.

TABLE IV
UTILITY FUNCTION ESTIMATES AND FUND ATTRIBUTE SUMMARY STATISTICS

Attribute	Utility weight, basis points (s.e.)	Mean	Standard deviation
Constant	93.53 (69.94)	N/A	N/A
Any load dummy	-12.11 (33.74)	0.547	0.499
Rear/deferred load dummy	59.57 (37.13)	0.272	0.446
Exchange-traded fund	199.5* (58.88)	0.023	0.149
Number other share classes	2.726 (9.722)	1.621	1.337
log[no. funds in same mgmt. company]	30.97* (12.67)	4.259	1.215
log[fund age]	99.39 (54.81)	1.393	0.728
Manager tenure (years)	3.578 (11.03)	2.922	2.776
Income + capital gains yield (percent)	-6.552* (3.009)	3.248	3.363
Avg. monthly percent diff. between fund and S&P 500 returns	136.4* (56.69)	-0.026	0.106
Standard deviation of monthly returns (percent)	48.22* (8.101)	4.455	1.293
N	309		
R ²	0.354		
Mean of dependent variable	582.9		

The table shows implied utility weights (β) of fund attributes using indirect utility levels implied by the differentiated-products model in subsections V.C and V.D. Standard errors are clustered by fund. Fund age is instrumented for using summary measures (fund counts and average attribute levels) of two other sets of sector funds: other funds managed by the same company (excluding the own-fund observation), if applicable; and those managed by other companies. Instruments vary by fund-year observation. Averages of all nonage attributes (i.e., those listed in the table above) are used in the instrument set. See text for details.

Figure IV. Read:

- In the differentiated-products model, search costs fall for most investors over time.
- The median implied search cost falls from about **1.5 basis points** in 1996 to about **0.2 basis points** in 2000.

- But search costs rise in the upper tail of the distribution, above roughly the **85th percentile**.

Table IV. Read:

- ETF status carries a large positive utility weight of about **199.5** basis points.
- The log number of funds in the same family has a positive weight of about **30.97** basis points.
- Tax exposure enters negatively at about **-6.55** basis points per percentage point.
- Better relative performance versus the S&P 500 enters positively at about **136.4** basis points.
- The generic load dummy is insignificant, but the rear/deferred-load dummy is positive at about **59.6** basis points.

Economic message. Search is not the whole story. Investors do value extra-portfolio attributes. At the same time, the positive rear/deferred-load coefficient is suggestive of switching-cost effects embedded in market shares.

5.6 Figure V: the upper-tail search-cost increase is concentrated among load-fund buyers

Read:

- Search costs are much lower for **no-load** investors than for **load-fund** investors.
- Over time, search costs fall for nearly all no-load investors.
- For load-fund investors, search costs rise for roughly the top **40%** of the distribution.

Economic message. This is the most direct evidence for the paper’s “novice investor” interpretation. The late-1990s shift toward expensive funds seems concentrated where one would expect information costs to be highest.

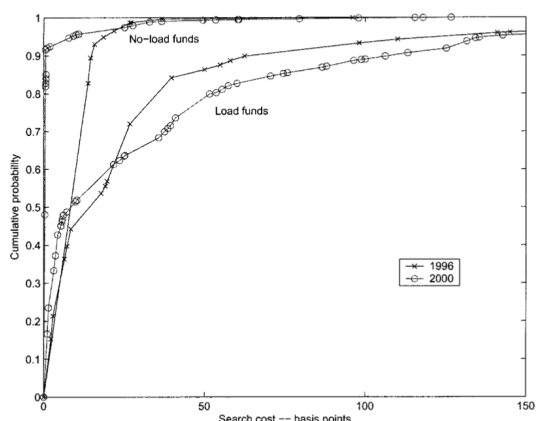


FIGURE V
Estimated Search Cost Distributions, 1996 and 2000
The figure plots the search cost distribution c.d.f.s obtained when the heterogeneous-funds model estimation in Figure IV is replicated after dividing investors into load fund buyers or no-load buyers. (See subsection V.E for details.) The plotted points are the nonparametrically identified cutoff search costs.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
		Assets	Search	Product	= \$1.22 m × [(1)-(1)]	= (2) × (3)	= (2) × (4)	= (5) + (6) + (7)	= (8) + (2)
Year	Funds	(\$billion)	savings (basis points)	variety cost (basis points)	Fixed costs savings (\$million)	Savings from search (\$million)	Product variety cost (\$million)	Net welfare change (\$million)	Indifference monopolist price change (basis points)
1995	24	22.0	65.1	-11.8	28.1	143.2	-26.0	145.3	66.1
1996	33	39.4	54.4	-13.0	39.0	257.7	-51.2	245.5	62.3
1997	44	70.6	44.7	-17.8	52.5	315.6	-125.7	242.4	54.3
1998	57	118.0	40.2	-23.1	68.3	474.4	-272.6	270.1	22.9
1999	68	174.8	42.5	-25.5	81.7	742.9	-445.7	378.9	21.7
2000	82	163.8	23.6	-30.8	98.8	386.6	-504.5	-19.1	-1.2

This table summarizes the welfare effects of a counterfactual industry structure where a Vanguard 500 Index Fund monopoly is imposed. Details on the calculations are found in the text and in the Computational Appendix.

5.7 Table V: welfare under a Vanguard monopoly counterfactual

Read:

- The monopoly counterfactual yields net welfare gains of about **145.3 million dollars** in 1995, **245.5 million** in 1996, **242.4 million** in 1997, **270.1 million** in 1998, and **378.9 million** in 1999.
- By 2000, however, the net welfare change is about **-19.1 million dollars**.
- The “indifference monopolist price change” is still large and positive through 1999, but slightly negative in 2000.

Economic message. Earlier in the sample, the search and scale-economy costs of fund proliferation appear to outweigh variety gains. By 2000, however, the product-variety benefit has become large enough to roughly offset those costs.

6 Short summary

The paper uses the retail S&P 500 index fund sector to study a broad industrial-organization question: how can many firms charge very different prices for what look like almost identical products? The answer is that these funds are not identical once one considers nonportfolio attributes and investor information frictions. A structural search model fitted to prices and market shares shows that both search costs and extra-portfolio product differentiation matter. Search costs are not uniform across investors and evolve over time: they fall for much of the population in the late 1990s but rise in the upper tail, especially among load-fund buyers. The paper interprets this as consistent with the entry of novice, high-information-cost investors. The broader message is that even in a market for nearly homogeneous financial products, competition is shaped by product positioning, visibility, and investor frictions rather than by portfolio returns alone.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the large number of mutual funds and the wide dispersion in their fees cannot be understood solely through portfolio heterogeneity. Even in the S&P 500 index-fund sector, where products are financially very similar, investors do not all end up in the lowest-cost fund. Search frictions and nonportfolio differentiation are central to market outcomes.

7.2 Economic intuition

An investor does not choose among funds in a frictionless spreadsheet. Finding, evaluating, and comparing funds is costly, and some funds are easier to notice than others. On top of that, investors value attributes such as family affiliation, tax properties, ETF structure, and perhaps advisory services. These features let multiple funds coexist and let higher-fee funds survive.

7.3 Takeaway

This paper demonstrates that “homogeneous financial products” can still generate rich industrial-organization patterns. The relevant dimensions of competition are not only returns and risk, but also search, visibility, distribution channels, and product design outside the portfolio itself.